

Institute of Actuaries of India

Subject SA6 – Investment

March 2017 Examination

INDICATIVE SOLUTION

Solution 1:**i) Impact of Demonetization**

We can use the framework provided by the Quantity theory of money to analyse the potential impact:

$MV = PY$, where

- M refers to the money supply
- V is velocity, the rate at which money turns over (the value of final sales [GDP] per rupee note)
- P, the price level
- Y, real GDP

There are two aspects to be analyzed in the current context. The first one is the reduction in money supply in the short run since the cash-based transactions would no longer be valid until old notes are replaced with new notes. However, on the other hand, the banks would see an increase in funds due to some unaccounted money being deposited back and therefore leading to an increase in money supply.

Demonetization first reduces the money supply - even genuine cash-based transactions cannot be carried out in the short run. However, it also has a secondary impact on velocity of money (V) since consumers are less likely to spend their precious cash (in new currency). As a result, there is a contraction in the velocity of money resulting in contraction in demand for goods and services, deferment of private capital expenditure, loss of employment, reduction in earnings power and further curtailment in spending. This will be most pronounced in the informal sector where transactions are purely cash based.

The velocity of money affects the money supply and interest rates and shifts the IS-LM curve resulting in contraction in demand and output. A corollary of this theory would be that interest rates might further decline. . This shift in the IS-LM curve would affect the demand for many goods including 2-wheelers, cars, consumer durables, gold, costly items like jewelleries as well as housing and other related industries like cement over the next few months / quarters.

In the short run, the circulation of money, which is like the blood flow, is slowing down, leading to a decrease in demand. When demand slows down, the production may slow down, employment may drop and investments may reduce. Thus there can be short term impact on business, jobs and income levels.

Specific comments on key inflation and interest rates are given below :

Points covering Inflation, Interest Rates, Taxation and Economic Growth

Reducing the base money supply will lower inflation (viz., essentially this is the opposite of too much money chasing too few goods). Of course, it's possible that the shock to the economy will lead to a reduction in production and the total effect will depend upon the interaction. If the destabilization reduces production by more than that fall in the money supply then we'll have more, not less, inflation. It would be extraordinary to believe that production is going to fall that much.

Then there is the taxation effect. At least some of that money will indeed be taxed and it would not have been taxed without the demonetization program. This means that the budget deficit will be smaller than it otherwise would have been. This will have knock on effects upon interest rates and inflation once again.

Consumer spending activity is expected to fall significantly. Consumers might refrain from making any purchases except essential items from the consumer staples, healthcare, and energy segments.

Activity in the real estate sector, which typically includes a lot of cash transactions is also expected to slow down significantly. This could further put downward pressure on asset prices.

Food item inflation, measured by changes in the Consumer Food Price Index, is also expected to fall since the supply and demand of food items would both get affected. Certain sources of food (e.g. fish markets) are completely cash based and these markets are expected to take a very strong hit. This will exert more downward pressure on inflation.

With the announcement of demonetization and the subsequent impact on cash transactions and consumption, inflation will likely fall. Banks will reduce their deposit rates. Lending rates are expected to fall as well. Due to the expected fall in inflation, the RBI will likely undertake more cuts in the repo rate.

These rate cuts will aim to boost consumption. They will work to get the economic growth back on track. A cut in interest rates boosts economic activity and raises hopes that after a lull for two to three quarters, economic growth may begin to rise.

Banks also expect to see a large rise in deposit accounts due to the demonetization. They are therefore expected to sharply reduced their deposit rates.

Loan burden will also fall - apart from cutting deposit rates, banks are expected to reduce their lending rates as well.

It will translate into lower interest rates on existing floating rate loans and new loans. The reduction in lending rates is expected to stoke lending by tempting consumers to take out loans for purchasing expensive consumer discretionary items like vehicles and houses.

Demonetization can help monetary transmission - Monetary policy transmission is the translation of monetary policy actions into the financial system through banking and trading channels. After being flushed with deposits, it's expected that banks will reduce their lending rates sizeably, improve the monetary transmission, and benefit consumers—both corporate and individual.

The lower interest rates may lead to more consumption and investment demand and thus gradual increase in economic activity which can be beneficial to India-focused funds.

(12)

ii)

- **Real Estate :**

- The impact may be felt more in the unorganized sector and secondary (resale) market involving cash transactions.

The property market can be affected by the government's decision to demonetize Rs. 500 and Rs. 1000 notes. The housing prices are expected to go down, fueling demand for the sluggish sector.

There may be a smaller impact on the organized primary residential market because the buyers in this category usually make purchases in the form of loans/mortgage or making cheque payments.

Commercial real estate is also expected to be impacted, because cash components do play a part in these transactions. However demonetization is being looked upon as a good move to institutionalize the real estate sector. One can expect more opportunities for institutional capital, debt investments, private equity and FDI players because of the increased transparency in the dealings. Banks can also start funding land and plot-based transactions, reducing land prices.

- **Government Securities and Corporate Bonds**

With the rupee remaining relatively stable, the debt market has reacted positively to the demonetization move.

Bond prices may remain elevated as the flight to safety would continue. This is because of the deflationary impact of demonetization; there is likely decline in inflationary pressures as demand comes down in the short term. This will also keep prices in check as the ability to hoard commodities and other assets will be greatly reduced.

Improvement in government finances due to shift of the black economy to white—increased tax compliance and better revenues for government— is another positive aspect.

The debt market has also started expecting further rate cuts, which, again, is good news for debt investors. With the household inflation expectations coming down, the possibility of rate cuts is increasing.

- **Infrastructure**

The infrastructure sector is driven by massive investments from government as well as loans from banks. One of the healthy effects of demonetization is to increase the liquidity in the banking system as well as increase the funds that the government has, for spending on items like infrastructure, welfare etc, so the Infrastructure is an obvious beneficiary of that.

However private infrastructure companies might suffer short term problems owing to wage payments, collections at toll booths etc. In the short run we do expect some negative sentiment prevailing in this sector.

- **Gold and Jewellery**

Generally the prices are likely to spike in the short term because gold is considered a safe haven. However it seems that the prices have given up the gains later. So, one can assume that the knee-jerk reaction is over now.

The demonetization move may have brought the money into the banking system and made people look at other investment avenues (rather than gold).

Jewellers are likely to face a short term fall in demand due to lack of cash based transactions.

- **Banking**

This is one sector, which is expected to benefit because a lot of money will be deposited in bank accounts.

The expectation is that a large part of the deposits would be made in Current & Savings Accounts thereby effectively making the net cost of deposits of these additional inflows

very low. This would tend to increase the freedom banks have to cut the lending rates without causing a major drop in their net interest margin.

In the short run – there could be lack of focus on lending, insurance and other services due to massive effort to cope with the demonetization drive. Overall however, in the medium to long term this sector must witness growth.

- NBFCs and Housing Finance Companies

NBFC and HFC generally grant loans to low-income workers but they should not face too much of an impact since these real estate sales are driven by “end-use” purpose and not “investment” purpose.

However rising competition from Banks (who would lower their interest rates given the inflow of funds) is expected to affect the profit margins for these companies. This might also affect the perceived value of the stocks (and might dampen demand for these stocks).

- Oil and Gas

This is an essential commodity – both for industrial and household consumption. The demand for these products are not expected to fall despite the demonetization drive. Therefore, the stock prices are not expected to get negatively impacted due to demonetization.

- Automobiles and Consumer Durables

This sector involves cash payments and is expected to be impacted.

The two-wheeler segment is more cash denominated (especially in rural areas) and hence it is likely to be impacted. Private passenger cars are not expected to be hit very hard (since cash transactions are not significant for this segment). Second-hand sales are more cash denominated and hence would likely to take a hit (especially commercial vehicles).

Many consumer durable giants like Nestle, HLL, ITC etc. are expected to see some decline in sales since cash would not be available for transacting, especially in semi-urban and rural areas.

- Pharmaceuticals

This is also an essential commodity. The demand for these products are not expected to fall despite the demonetization drive. In the immediate near-term, demonetization could have an impact whereby only critical medicines are bought and expenditure on supplements, vitamins, health drinks etc. drops. However, in the medium to long term there should not be a significant impact on the demand for these products. Therefore, the stock prices are not expected to get negatively impacted due to demonetization.

- Listed Online Mobile payments company

There should be an uptake in demand for using the services of these Companies given the government initiative to curb black money and bring in transparency. These Companies are therefore expected to have huge growth potential and consequently the stock prices of these companies are expected to grow steeply.

Further due to the capping of withdrawal amount per week it is expected that people would be forced to use the services of these companies – increasing the business for these firms.

(20)

iii)

For the sectors that are expected to get hit:

- For sectors where the expectation is that the share price would rise
 - Go Long on Futures for Government Bonds
 - Go Long on Futures for Banking stocks and others that are expected to be positively benefited
 - Buy Calls for Banking stocks and others that are expected to be positively benefited
 - Option strategies with better pay-off on gains rather than fall.
 - Bull Spreads
- For sectors where the expectation is that the share price would fall
 - Short the Futures with exposure to the assets/sectors where prices are expected to drop)
 - Buy puts (explain briefly what this means)
 - Create option strategies which give a larger payoff on fall rather than gain :
 - Bear Call Spread

- Bear Put Spread
- Strap

(6)

iv) - In the long term, demonetization will help the mutual fund industry. The banks will reduce their fixed deposit rates further. So people will start looking at mutual funds to park their surplus funds to earn inflation-beating returns.

- With stock markets being volatile due to demonetization of currency, one can invest in good quality mutual funds.

- As people will deposit money in banks, overall savings will increase, which is expected to have positive impact on money inflow to mutual funds

-In the fixed income side the yields are falling which will benefit the debt funds in the short term. With more cash coming into the overall banking system and the mainstream the overall impact on equities will be positive in the long term.

(2)

[40 Marks]

Solution 2:

i)

Liability Driven Investment (LDI) is the terminology used to describe an investment methodology wherein the asset allocation, either in part or in whole is determined by taking into account (i.e. relative to) the specific set of liabilities.

LDI is not a strategy or a type of product available in the market but an approach to setting the investment strategy.

The approach is commonly used by insurance companies and defined benefit pension funds to manage the mismatch between their assets and liabilities (which comprise an income stream to annuitants).

Under an LDI approach it is possible to closely match:

- the interest rate sensitivity (duration) of the liabilities
- the inflation-linkage of the liabilities
- the shape of the liabilities

Interest rate risk

The present value of fixed-rate cashflows payable in the future is linked to the interest rate used to value them. As interest rates rise, the value of fixed rate liabilities fall and vice versa. The greater the length of time (or duration) until the future cashflows are due to be paid the more sensitive the value is to a change in interest rates. Interest rate risk can be reduced by investing in instruments which match the duration and value of the fixed rate cashflows payable. Investments that are used to match duration include fixed rate bonds and interest rate swaps.

Interest rate swaps are typically provided by investment banks as an over-the-counter product. It carries with it additional credit risk. A typical “receive fixed pay floating” interest rate swap can hedge the risk that future interest rates might fall.

Inflation linked risk

If an investor has liabilities linked to inflation, their present value will be sensitive to changes in inflation expectations. An investor would need to invest in assets with the same sensitivity to inflation expectations as the liabilities to reduce any mismatch in performance. Investments used to match inflation liabilities include inflation linked bonds and inflation swaps.

Inflation swaps are typically provided by investment banks as an over-the-counter product. It carries with it additional credit risk. (Additional marks for describing an inflation swap)

The shape of the liabilities

The shape of the liabilities will depend on when the cash-flows are expected to be paid. For many investors the most straightforward way to match the liabilities would be through holding a portfolio of bonds. Although it is possible to construct a bond portfolio where bond payments match the projected liability payments for a pension fund it is often more difficult to match longer duration payments (40-50 years) due to the limited issuance or non-availability of bonds in some countries. This presents particular challenges for long-dated liabilities, especially inflation linked liabilities. In order to purchase assets that match the shape of cashflows at longer durations, investors rely on using swaps to hedge both interest rate and inflation risks. More complex hedging portfolios may also make use of repo transactions to hold some of the bonds on an unfunded basis.

In addition to interest rate or inflation hedging, a number of pension funds and insurance companies have also entered into longevity swaps or longevity insurance policies that exchange fixed payments (“premiums” or expected payments to annuitants).

(12)

ii) Schemes that have not used matching strategy will have higher proportion of assets invested in domestic and overseas equities and property and less in gilts and corporate bonds.

There may be various reasons for having a bond based strategy over years:

The long term interest rates would have fallen.

The return on equities and property might have been lower as compared to the bonds.

The availability of long dated bonds would have increased due to demand from insurance companies and pension funds.

May be due to regulatory changes that would have emphasized on de-risking.

Repairing of deficit which is done either by additional contributions to the scheme or by adding a lumpsum with aim of buying out benefits is easier when the position is matched.

This is due the fact the volatility in equities can make difference in funding situation year on year especially in case the equities are not doing good as asset class. It becomes difficult to estimate the future fundings.

Buy out prices may be linked to bond rates, which makes a case for having a bond based matched strategy.

(5)

iii) A number of approaches can be used to reduce or eliminate duration risk. These include:

- conventional approaches
- using repos
- using swaps
- using swaptions
- pooled liability-driven investment funds.

(3)

iv)

Swaps:

Swaps are flexible in terms and can be customized as per needs of the scheme. The scheme can accurately design the exact duration it wants.

However there is counterparty risk in a swap. This can be reduced by margin or collateral but it still is a potential problem.

Swaps are illiquid and may cause operation or valuation problems in the future.

A large swap portfolio can cause problems in the future if the scheme is split through mergers or acquisition.

Long gilt futures:

This would be quick and easy to place. Since it is mostly exchange traded there is more security in entering the transaction. If the future is available for longer terms then it may hedge the interest rate risk in the future.

However there may be substantial margin requirement for large future which may give rise to cashflow problems.

The future generally has short duration to expiry, hence the future needs to be rolled over indefinitely. This would rise to basis risk.

If the underlying bond is not of a long duration then it will not be effective at hedging changes in interest rate at longer duration.

Swaptions:

Swaptions allow scheme the right to enter into a fixed/floating swap at a point in the future, for example a time when a personal pension vests.

An institution with duration risk to falling interest rates might undertake a swaption to receive a fixed rate of say 8% and pay floating over a future period.

In this way, it can effectively guarantee to receive an interest rate of at least 8% pa.

Since the swaption's value will increase as the market swap rate falls, these profits will offset the rising liability value. The benefit is that if interest rates rise, the swaptions can only fall to zero value, and cannot become a liability as happens with a swap.

This can leave the institution with a net gain in a rising interest rate scenario. However, the gain would have to be weighed against the cost of the swaptions.

In other words, and just as with vanilla put and call options on shares, swaptions have limited downside risk. Ignoring counterparty risk, the most that can be lost is the initial premium paid.

(8)

v) ETFs are funds that are exchange traded like normal shares.

They are a mix between the traditional collective investment schemes – investment trusts and unit trusts.

There are a large number available in the markets, and they are linked to all sorts of indices, including equity indices, value/growth shares, sector indices, gold, oil, and bonds.

The charges on ETFs are considerably less than those on unit trusts and investment trusts.

Although they trade on an exchange, the price does not trade at a discount (or premium) to the net asset value, like an investment trust.

This is because the mechanism allows large scale market players to trade with the manager of the ETF, and ask to exchange shares for the matching portfolio, or vice versa.

If the price falls below the net asset value, a market trader may ask to give a large portfolio of the matching assets to the company that manages the ETF in exchange for a bundle of new shares in the ETF.

This will push the price of the shares back up to the NAV and avoid the discount that affects investment trusts.

Only very large traders can do this and the scheme would not be able to obtain shares through this route.

The shares available on the secondary market are generally not available in the size that the scheme would be considering, so it would be difficult to obtain sufficient quantities of shares in the chosen ETF.

In general they are tracking funds, and so do not offer active management. This might be a negative for the scheme.

(5)

vi) The key objectives generally relate to the control of features such as:

(a) future ongoing funding levels. This is often viewed by actuaries as the best all-round, long-term measure of the health of the pension fund.

(b) future solvency levels including scheme-specific definitions of solvency or, more probably, MFR funding levels (even though the MFR is not strictly speaking a test of solvency). This is a shorter-term objective than one concentrating on the ongoing funding level. Controlling solvency levels or MFR funding levels may therefore require a different strategy to that which might be ideal for controlling ongoing funding levels. Trustees are likely to be particularly interested in the future position of the scheme vis-a-vis the MFR since they will have to comment on it in the statement of investment principles.

(c) future company contribution rates or accounting measures of the cost of providing pension benefits. Asset/liability studies often include an analysis of pension costs, because this is the measure of pension cost that impacts on the sponsoring company's bottom line.

(3)

vii) Steps in an ALM exercise:

1. The key objectives that investment and funding policy should aim to achieve need to be clarified. These involve objectives such as:

- future on-going funding levels
- future solvency levels
- future company contribution rates
- the level of risk (performance mismatch between assets and liabilities) that is prepared to be taken

2. Suitable assumptions to use in the study need to be agreed.

3. Data on the liabilities needs to be collected to carry out the projections. For detailed liability analysis data on individual members is required to build up an accurate assessment of the future cashflow projections.

4. The overall nature of the liabilities is considered — an analysis of current funding level, maturity and cash flow projections under different scenarios is considered.

5. An analysis would be carried out to identify how the pension fund might progress in the future if different investment strategies were adopted.

6. Different asset mixes would then be analysed in more detail to assess the risks (relative to the liabilities) and the rewards of each alternative under consideration.

7. The results would be summarised and presented. Compared to a traditional actuarial valuation, an ALM provides much more information in three (or more) extra dimensions:

- providing projections into the future (time dimension)
- providing some estimate of the range of likely outcomes (probabilistic dimension)
- indicating the effect of changing investment strategy (asset mix dimension)

In steps 5 and 6, ALMs use an asset model to produce stochastic simulations of returns on asset classes and other relevant economic data (such as inflation). For pension funds looking to hedge their liabilities or adopt an LDI approach to investment strategy an in-depth analysis on the impact of changes to the liability profile is undertaken. The output from the model is used

to calculate (approximate) liability values at different time horizons. The models in common use are all designed to be used in Monte Carlo simulation exercises.

(5)

viii)

The key fundamental assumptions are the ones concerning the future behavior of economic variables that influence how the scheme's assets and liabilities might change in the future. In totality these assumptions are generally referred to as the stochastic investment model underlying the study. For a pension fund asset/liability study the stochastic investment model typically involves assumptions about the future behavior of the following variables:

- a) price inflation;
- b) salary growth; and
- c) returns on assets (and yields on assets, where relevant).

In addition to assumptions which make tractable the projections, it is also necessary to make assumptions regarding:

- (a) investment policy, i.e. the choice of investment policy to analyse and whether they should be limited merely to static ones or also include dynamic ones;
- (b) economic, i.e. the stochastic investment model underlying the projections;
- c) methodology, e.g. what constitutes solvency and what key features need to be projected;
- (d) demographic, e.g. what are suitable assumptions for retirements and deaths; and
- (e) funding and valuation policy, e.g. how contribution rates will progress in the future, and how ongoing actuarial valuations will be carried out.

(4)

ix) The common aim of asset/liability studies is to review the investment policy. However it may also be helpful in providing insights into:

Future ongoing funding levels which is viewed as the best long term measure of health of the pension fund.

Asset liability studies often include an analysis of pension costs, because this is the measure of pension costs that impacts on the sponsoring company's bottom line. The ALM studies help in assessing the future company contribution rates or cost of providing pension benefits.

ALM studies also help in providing scheme specific future solvency levels. It also provides estimate of minimum funding levels in future. Trustees may in particular be interested in future position of schemes vis-à-vis the MFR.

ALM studies are also used to test the impact of changing funding policy.

(3)

x) The answer is that an asset/liability study provides much more information than is available from an actuarial valuation. A valuation provides a single 'answer' at a set point in time (the valuation date). In contrast, an asset/liability study provides three or more extra dimensions by:

- (a)** providing projections into the future (introducing a time dimension);
- (b)** providing some estimate of the range of likely outcomes (a probabilistic dimension);
and
- (c)** indicating the effect of changing investment strategy (an asset mix dimension).

(3)

xi) The conventional method used to identify 'optimum' portfolios involves defining:

- (a)** a measure of 'reward' or 'return' from each asset category
- (b)** a measure of 'risk' from each asset category, and
- (c)** correlations between different asset categories, so that the returns and risks of mixtures of assets can be identified.

Constraints are also placed on portfolios. Typically portfolios are required not to be geared or short-sold in any market. Constant asset mixes are usually assumed.

Any acceptable portfolio can then be categorised by its 'return' and 'risk', measured as per (a) and (b).

For any given level of return there will be a portfolio (or occasionally more than one portfolio) with the minimum possible risk. Or, equivalently, for any given level of risk there will be a portfolio which has the greatest possible return.

Portfolios with these characteristics are described as efficient portfolios. These will range from one with the lowest possible risk to one with the highest possible return (if as is usual our measures of risk and return are such that increasing risk is rewarded with extra expected return).

The whole series of efficient portfolios each achieving an optimal trade off between risk and reward (for some given risk) is known as the efficient frontier.

Then the investors utility function (usually expressed as a function of expected return and variance) is plotted against the efficient frontier. The point of intersection is the optimum portfolio for the investor based on the pre-specified assumptions.

(5)

xii) The following are the implications:

1. The two tests under Ind AS 109 – Business Model Test and Contractual Cashflow test – need to be applied here to arrive at the classification and measurement of the debt instruments concerned.
2. Previously, these debt instruments would have been measured at amortized cost since they were previously classified as “Held till maturity (HTM)”.
3. Now these would be measured as HTM, or FVTOCI (fair value through Other Comprehensive income) or Held for trading or a mix of the three, depending on the classification the company chooses to go with, which in turn will depend on how frequently the assets are transacted/traded.
4. Many companies may choose the FVOCI route wherein under Ind AS 109, for these debt instruments, interest income and impairment (along with exchange gains/losses for foreign currency denominated assets) will be recognized in the P&L statement and the changes in the fair value of the debt instruments being routed through OCI (other comprehensive income).
5. The question does not mention explicitly that the trustees have elected to classify the debt instruments as FVTPL (fair value through P&L) and hence this is not explored further in the solution.
6. The difference between fair value of assets and amortized cost would be recognized through OCI. So if the fair value is higher than amortized cost then this would be recognized as an OCI gain. If vice-versa, it would be recognized as an OCI loss.
7. It would be useful to share the impact of such a shift on the balance sheet and projected profit and loss statement to the trustees. The following are the effects of the shift :

Item	Before reclassification as AFS	After reclassification as AFS	Remarks
P&L measurement	Effective Interest Rate * (Amortized Book Value as at the Opening date)	Same	If a Debt instrument is measured at FVTOCI – then the amounts recognized in the P&L equal the same amount had these instruments been classified using the Amortized cost method. (Para 5.7.11)
B/S measurement	Amortized Book Value shown as carrying amount	Fair Value shown as carrying amount with changes routed through OCI.	Market Fluctuations are now effectively accounted directly in the Balance sheet. Difficulties in obtaining fair value for certain debt instruments could pose a challenge. Amortized cost method usually has smooth recognition in the Balance Sheet until date of maturity or earlier redemption.

(4)

[60 Marks]
